

Global Health Policy

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DEFINITION:, 4, 6–8

FACT:, 4, 7, 8

A, 4

B, 4

Blue, 3

C, 4

Global Burden of Disease and Risk Factors (2013), 3

Global Burden of Disease and Risk Factors in Japan (2013), 13

Global Health Policy, 1

Green, 3

Influenza, 8

Measles, 8

Red, 3

Smallpox, 8

transmissibility, clinical seriousness, and population immunity
level, 7

1 Global Health Policy and Governance

1.1 Global Health Transition

Top 10 Causes of Death are:

Global Burden of Disease and Risk Factors (2013)	
Rank	Deaths
1	Ischemic heart disease (缺血性心脏病)
2	Cerebrovascular disease (脑血管疾病)
3	COPD (Chronic Obstructive Pulmonary Disease) (慢性阻塞性肺疾病)
4	Lower respiratory infections (下呼吸道感染)
5	Alzheimer disease
6	Lung Cancer
7	Road injuries
8	HIV/AIDS
9	Diabetes
10	Tuberculosis

Blue Non-communicable diseases
Red Communicable diseases
Green Injuries

The comparison of top 10 causes of death and disabilities are:

Global Burden of Disease and Risk Factors (2013)		
Rank	Deaths	Deaths and Disabilities
1	Ischemic heart disease	Ischemic heart disease
2	Cerebrovascular disease	Cerebrovascular disease
3	COPD (Chronic Obstructive Pulmonary Disease)	Lower respiratory infections
4	Lower respiratory infections	Lower back/neck pain
5	Alzheimer disease	COPD (Chronic Obstructive Pulmonary Disease)
6	Lung Cancer	Diarrhea
7	Road injuries	Road injuries
8	HIV/AIDS	Neonatal/Preterm births
9	Diabetes	HIV/AIDS
10	Tuberculosis	Malaria

Mortality and non-fatal health outcomes can be combined in a graph:

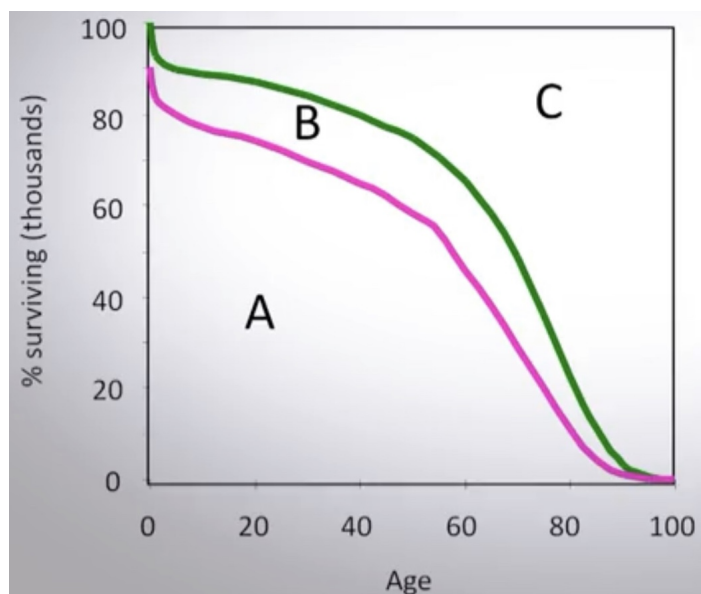


Figure 1. Lesson 1-1 3:04.

The **green line** represents the life expectancy at birth.

The **pink line** represents the proportion of people that are alive and healthy (without disabilities).

Area **A** is the proportion of people that are healthy.

Area **B** is the proportion of people that are alive with various disabilities.

Area **C** is the proportion of people dead.

1.2 Epidemiological Transition

FACT: Life expectancy at birth: 71.5 years (2013), 6.2 years increase from 1990.

FACT: Healthy life expectancy at birth: 62.3 years (2013), 5.4 years increase.

FACT: The gap between life expectancy and healthy life expectancy has increased.

DEFINITION: DALYS: Disability-adjusted life years.

The distribution of DALYS in various countries at different times:

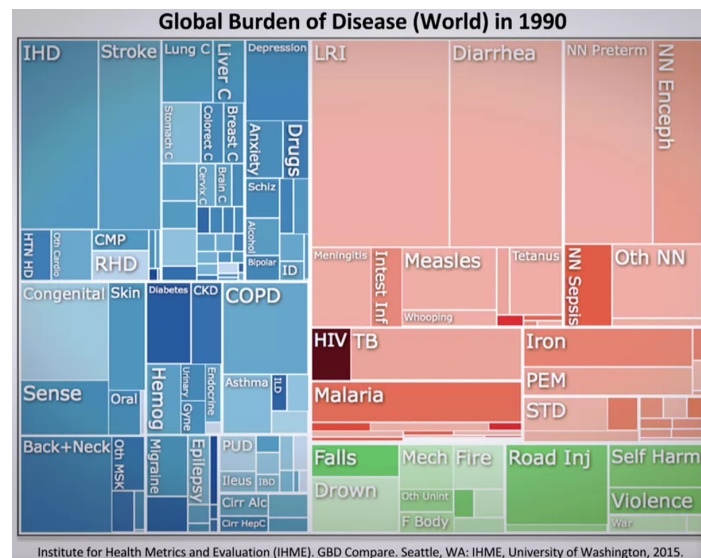


Figure 2. Lesson 1-2 3:34.

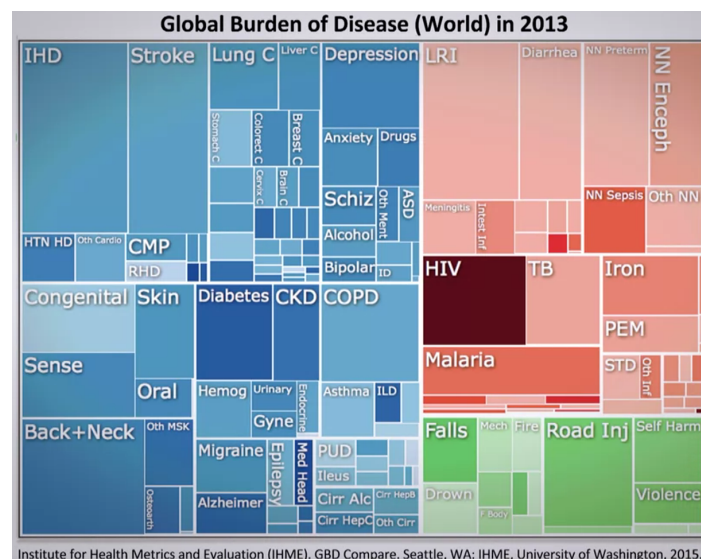


Figure 3. Lesson 1-2 4:22.

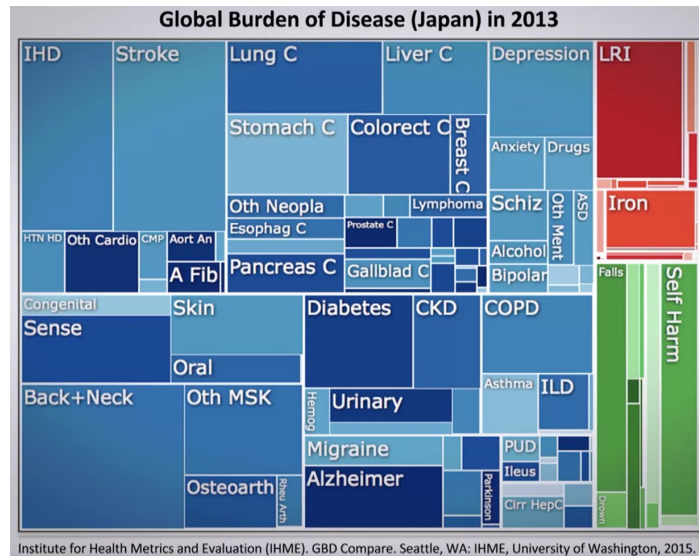


Figure 4. Lesson 1-2 4:22.

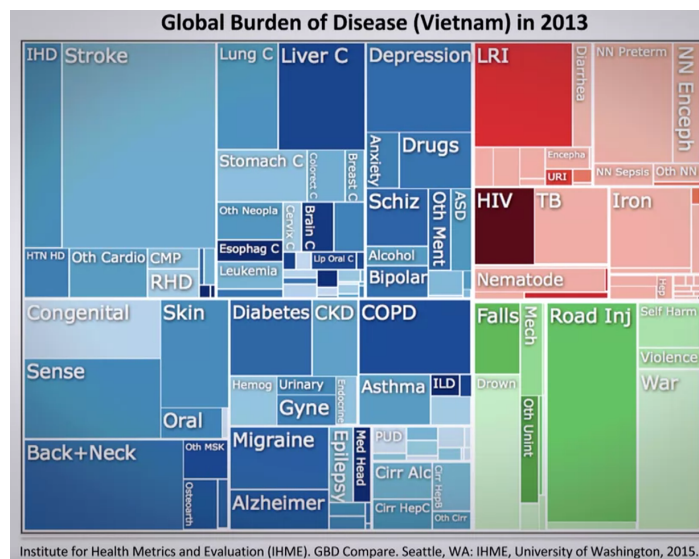


Figure 5. Lesson 1-2 4:22.

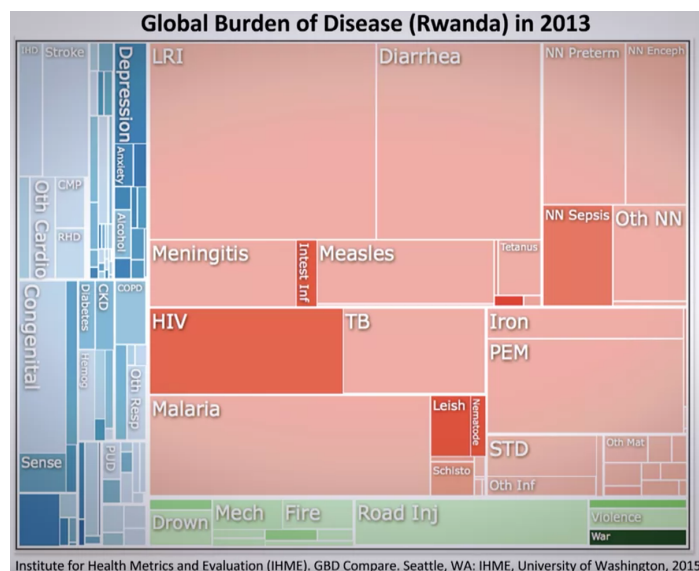


Figure 6. Lesson 1-2 4:22.

1.3 Evolution of Global Health

1.4 Global Health Governance

1.5 Global Health Challenges

1.6 Challenges in Global Health Policy

1.7 Japan's Commitment to Global Health

1.8 Proposals for the G7 Summit

2 Pandemics and Health Security Concerns

2.1 Pandemics and Health Security Response I

DEFINITION: Epidemic: Occurrence of infectious diseases in excess of the expected level

DEFINITION: Outbreak: Abnormal increase in the number of cases in a group of hosts

DEFINITION: Pandemic: Epidemic spread over several countries or continents

The historical occurrence of influenza pandemic timeline:

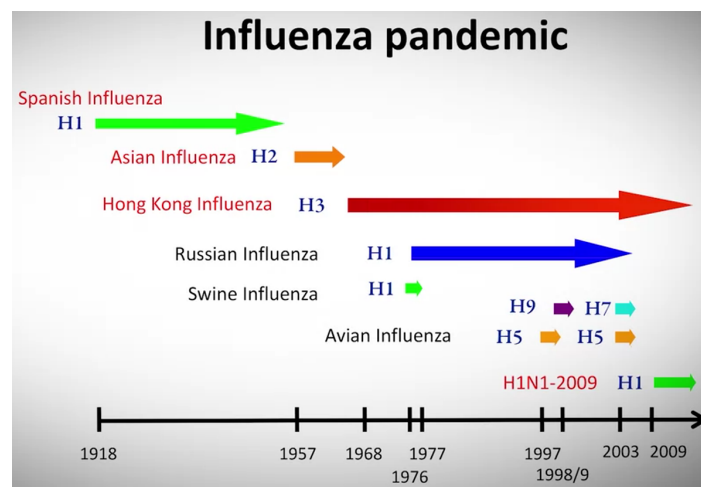


Figure 7. Lesson 2-1 3:04.

The location and deaths of influenza pandemic:

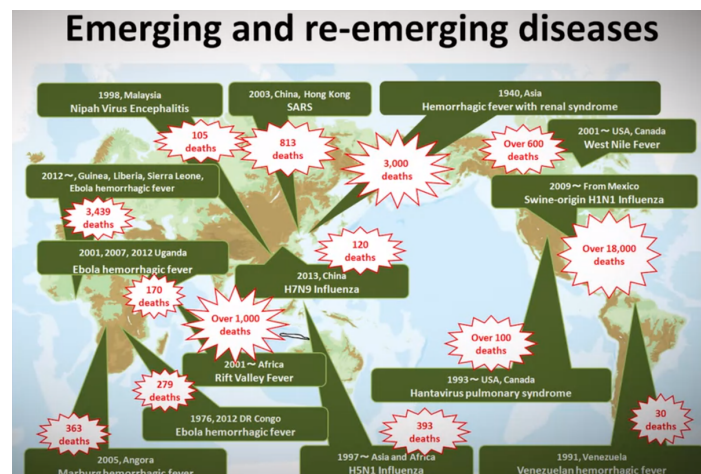


Figure 8. Lesson 2-1 4:10.

2.2 Pandemics and Health Security Response II

DEFINITION: Zoonosis: An infectious disease spread from vertebrate animals to humans

The barriers in the spread of infectious diseases:

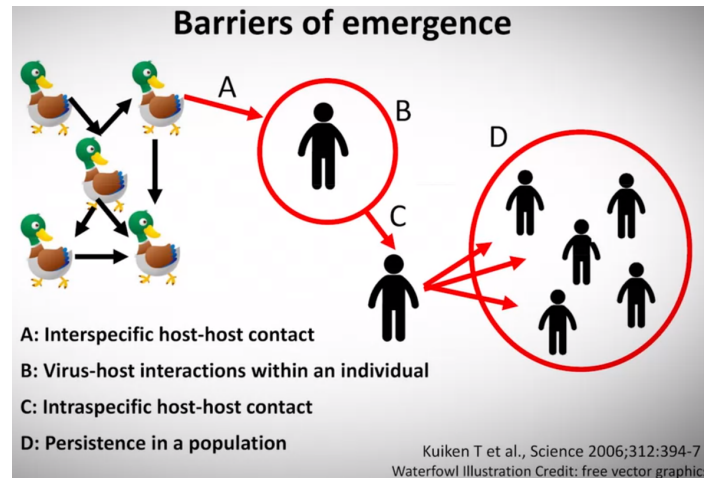


Figure 9. Lesson 2-2 7:22.

FACT: Zoonotic-origin infectious disease is becoming more common.

2.3 Outbreak Investigation "Consider Your Action" I

The steps in response to an outbreak investigation:

1. Establish case definitions:
Collect common sample of features
2. Confirm the accuracy of cases
3. Establish background rate of disease
4. Find cases, determine outbreak, decide outbreak scope
5. Examine descriptive epidemiologic features
6. Generate and test hypothesis
7. Collect and test environmental samples
8. Implement control measures
9. Interact with press and inform the public

2.4 Outbreak Investigation "Consider Your Action" II

The disaster size of a pandemic is determined by **transmissibility, clinical seriousness, and population immunity level**.

DEFINITION: Transmissibility, or transmission potential, is the percentage of population that will experience infection.

Transmissibility can be measured in confined space, such as in a household or in a classroom, with the measurement of secondary attack risk (SAR). It can also be measured in an open space, with the measurement of the basic reproduction number, R_0 .

DEFINITION: SAR measures the proportion of individuals that become ill among all exposures.

$$SAR = \frac{\# \text{ of ill}}{\# \text{ of exposures}}$$

DEFINITION: The basic reproduction number, R_0 is the average number of secondary cases generated by a single primary case.

FACT:

$R_0 > 1$ leads to a major epidemic.

$R_0 < 1$ declines to extinction.

FACT:

Influenza has an R_0 of 1.2 - 2.0.

Smallpox has an R_0 of 5.0.

Measles has an R_0 of 20.

FACT: Vaccination threshold can be calculated using:

$$R_v = (1 - p)R_0 < 1$$

$$1 - p < \frac{1}{R_0}$$

$$p > 1 - \frac{1}{R_0}$$

where p is the vaccination coverage,
and R_v is the reproduction number under vaccination.

2.5 Lessons from Pandemic H1N1-2009

Two possible interventions for pandemic is

DEFINITION: Containment: stop transmission at a geographic scale. (isolation, quarantine including border control, contact tracing)

DEFINITION: Mitigation: slow down the spread and minimize social impact (school closure, wearing masks and antiviral treatment)

2.6 Real-time Research Response

2.7 Disasters, health and human security

The causes of death in Japan 2010:

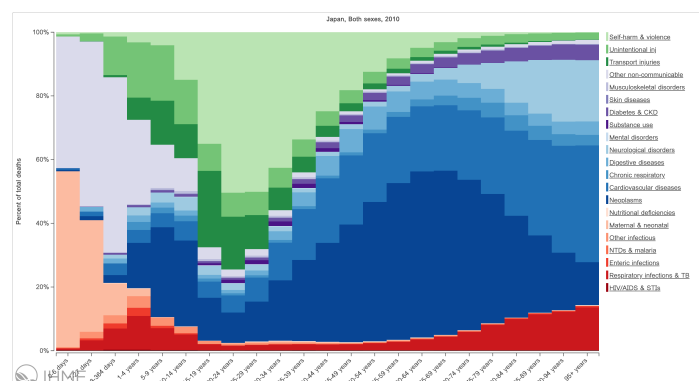


Figure 10. Lesson 2-8 2:43.

The risk factors of death in Japan 2010:

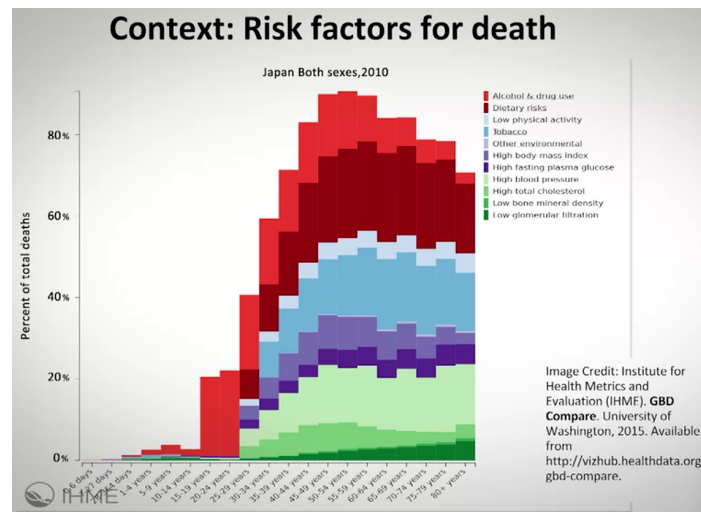


Figure 11. Lesson 2-8 4:23.

2.8 Acute effects

2.9 Long-term health effects

2.10 Long-term health effects

2.11 Health, human security and disaster response

3 Universal health coverage in the context of aging

3.1 Overview and concepts

The three measurements of universal health coverage is:

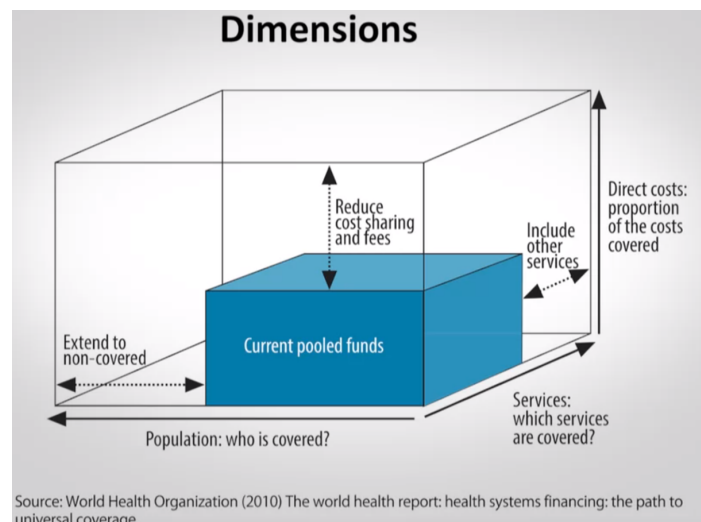


Figure 11. Lesson 3-1 5:07.

3.2 Financial Risk Protection I

3.3 Financial Risk Protection II

3.4 Health Service Coverage

3.5 Equity

3.6 Aging

3.7 Case study 1: Japan

3.8 Case study 2: Bangladesh

3.9 New Policy Directions

4 Universal health coverage in Japan

4.1 UHC in Japan: Introduction

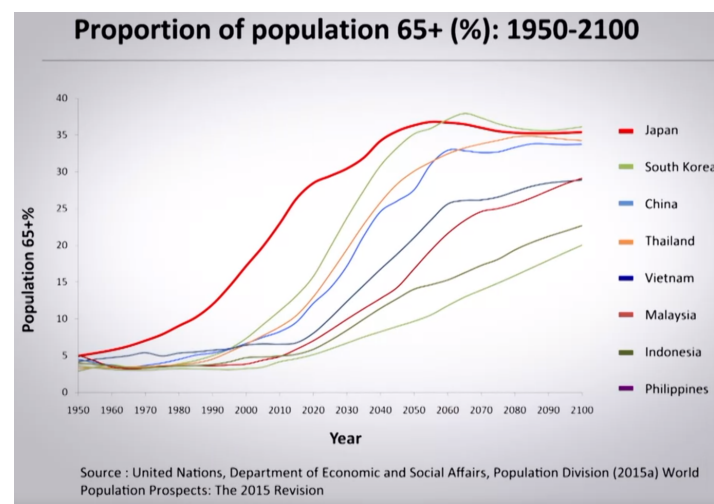


Figure 12. Lesson 4-1 5:56.

4.2 Understanding Health System Performance

4.3 Population Health

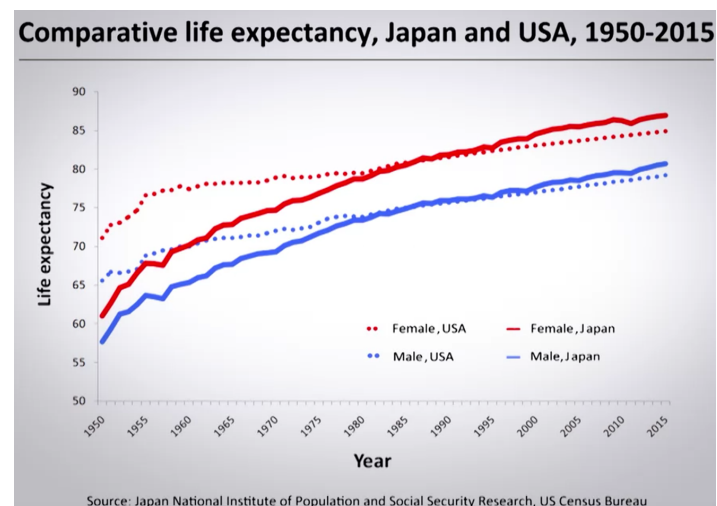
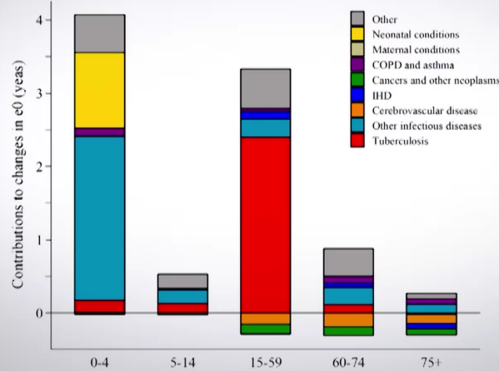


Figure 13. Lesson 4-3 0:15.

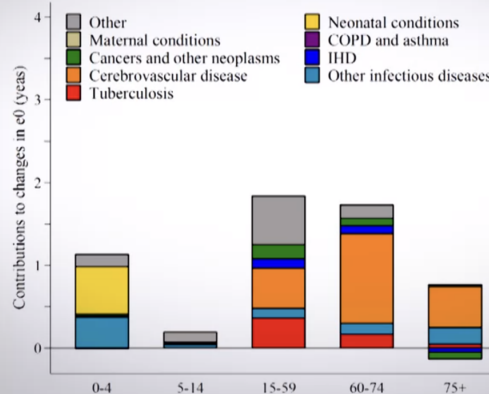
Age- and cause-specific contributions to changes in life expectancy at birth in Japan: 1950–1965 (males)



Source: Ikeda, N., Saito, E., Kondo, N., Inoue, M., Ikeda, S., Satoh, T., ... & Noda, M. (2011). What has made the population of Japan healthy?. The Lancet, 378(9796), 1094-1105.

Figure 14. Lesson 4-3 0:52.

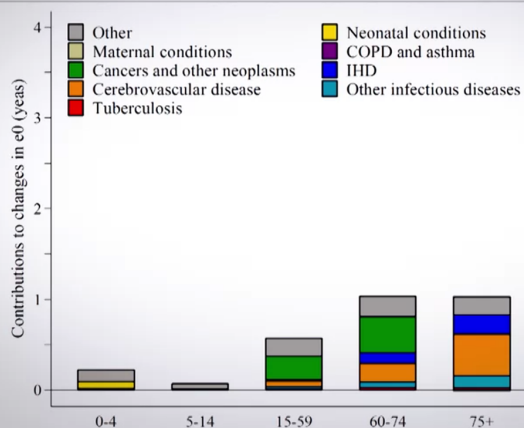
Age- and cause-specific contributions to changes in life expectancy at birth in Japan: 1965–1980 (males)



Source: Ikeda, N., Saito, E., Kondo, N., Inoue, M., Ikeda, S., Satoh, T., ... & Noda, M. (2011). What has made the population of Japan healthy?. The Lancet, 378(9796), 1094-1105.

Figure 15. Lesson 4-3 2:37.

Age- and cause-specific contributions to changes in life expectancy at birth in Japan: 1995–2008 (males)



Source: Ikeda, N., Saito, E., Kondo, N., Inoue, M., Ikeda, S., Satoh, T., ... & Noda, M. (2011). What has made the population of Japan healthy?. The Lancet, 378(9796), 1094-1105.

Figure 16. Lesson 4-3 4:30.

Age-standardized cancer mortality (males)

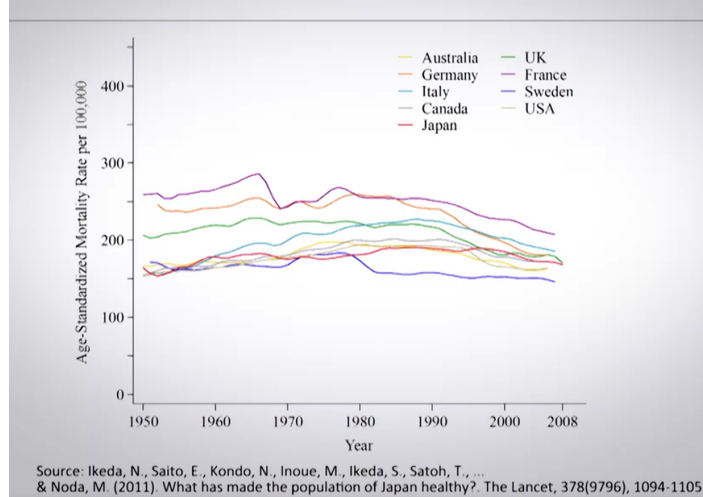


Figure 17. Lesson 4-3 3:10.

Age-standardized ischemic heart disease mortality (males)

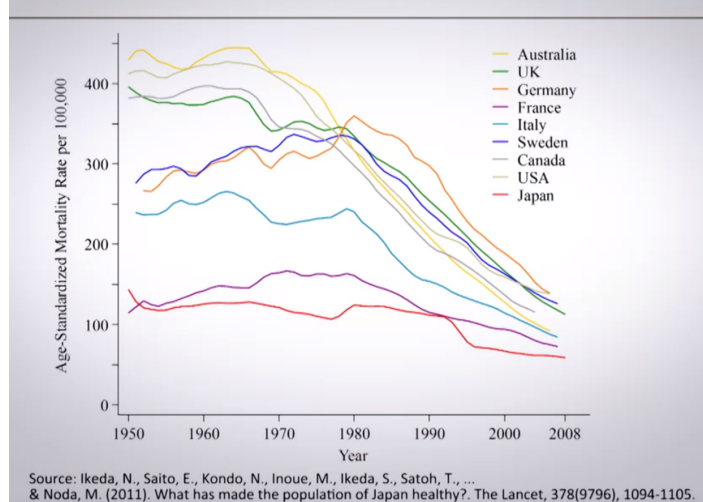


Figure 18. Lesson 4-3 3:38.

Age-standardized stroke mortality (males)

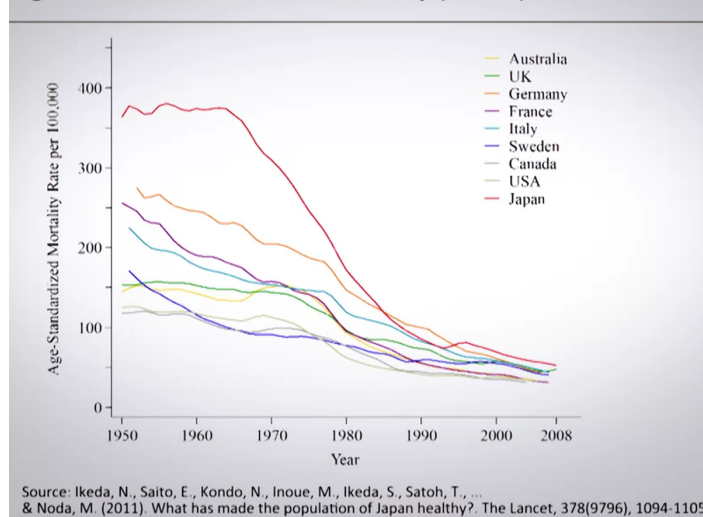


Figure 19. Lesson 4-3 3:40.

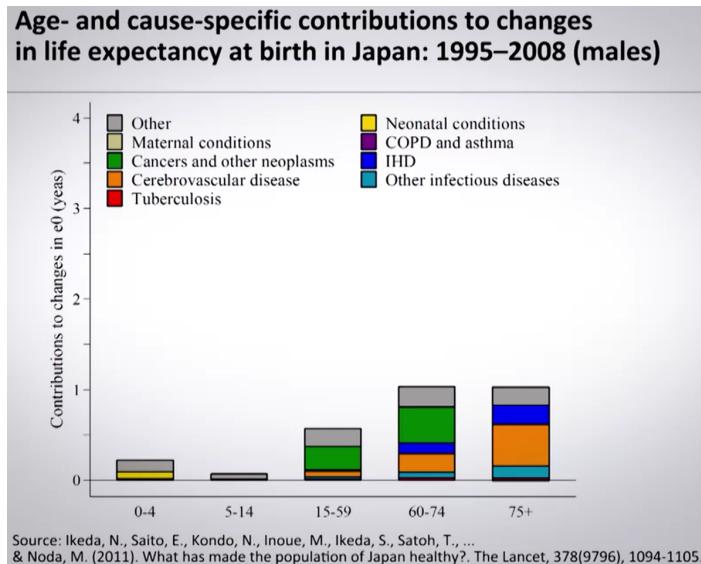


Figure 20. Lesson 4-3 4:30.

Global Burden of Disease and Risk Factors in Japan (2013)

Rank	Deaths	DALYS	
		Causes	Risk Factors
1	Cerebrovascular disease	Lower back and neck pain	Dietary risks
2	Lower respiratory infections	Cerebrovascular disease	High blood pressure
3	Ischemic heart disease	Ischemic heart disease	Tobacco smoke
4	Lung cancer	Lower respiratory infections	High fasting glucose
5	Stomach cancer	Lung cancer	Alcohol and drug use
6	COPD	Other musculoskeletal	High BMI
7	Colorectal cancer	Diabetes	Low physical activity
8	Alzheimer disease	Sense organ diseases	Low GFR (Glomerular Filtration Rate)
9	Liver cancer	Depressive disorders	Air pollution
10	Chronic kidney disease	Self-harm	High total cholesterol

Annual change in adult mortality rate (females)

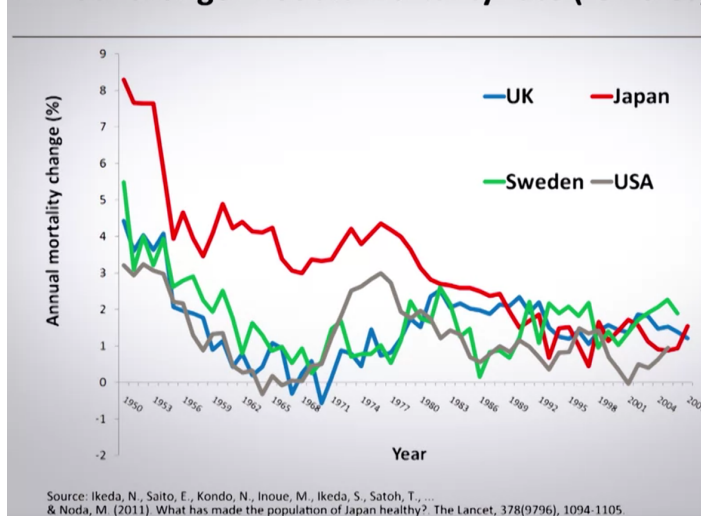


Figure 21. Lesson 4-3 5:29.

- 4.4 UHC in Japan: Achievements and Challenges**
- 4.5 Cost Containment**
- 4.6 Service Quality**
- 4.7 Population Aging**
- 4.8 Future Directions**